

14.8 Lagrange Multipliers

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In this section we present Lagrange's method for maximizing or minimizing a general function $f(x, y, z)$ subject to a constraint of the form $g(x, y, z) = k$.

Stewart explains the geometric basis for Lagrange's method on p. 957 and p. 958 of his book quite brilliantly, and you are strongly recommended to read through it. You might have to read up some terminology used in section 14.4 on tangent planes to understand his intuitive argument. We shall now briefly look at how this argument can be made precise.

Suppose that a function f has an extreme value at a point $P(x_0, y_0, z_0)$ on a surface S , resulting from the equation $g(x, y, z) = k$, and let C be a curve with vector equation $\mathbf{r}(t) = \langle x(t), y(t), z(t) \rangle$ that lies on S and passes through P .

If t_0 is the parameter value corresponding to the point P , then $\mathbf{r}(t_0) = \langle x_0, y_0, z_0 \rangle$. The composite function $h(t) = f(x(t), y(t), z(t))$ represents the values that f takes on the curve C .

Since f has an extreme value at P , it follows that h has an extreme value at t_0 , so, by Fermat's Theorem, $h'(t_0) = 0$.

But if f is differentiable, we can use the Chain Rule to write

$$\begin{aligned} 0 &= h'(t_0) \\ &= f_x(x_0, y_0, z_0) x'(t_0) + f_y(x_0, y_0, z_0) y'(t_0) \\ &\quad + f_z(x_0, y_0, z_0) z'(t_0) \\ &= \nabla f(x_0, y_0, z_0) \cdot \mathbf{r}'(t_0) \end{aligned}$$

This shows that the gradient vector $\nabla f(x_0, y_0, z_0)$ is orthogonal to the tangent vector $\mathbf{r}'(t_0)$ to every such curve C .

But (as you should verify) from Section 14.6 we know that the gradient vector $\nabla g(x_0, y_0, z_0)$ is also orthogonal to $\mathbf{r}'(t_0)$ for every such curve.

This means that the gradient vectors $\nabla f(x_0, y_0, z_0)$ and $\nabla g(x_0, y_0, z_0)$ must be parallel.

Thus, if $\nabla g(x_0, y_0, z_0) \neq \mathbf{0}$, then there exist a $\lambda \in \mathbb{R}$ such that

$$\nabla f(x_0, y_0, z_0) = \lambda \nabla g(x_0, y_0, z_0).$$

The number λ is called a **Lagrange multiplier**. The procedure based on the above equation is as follows.

Method of Lagrange Multipliers. To find the maximum and minimum values of $f(x, y, z)$ subject to the constraint $g(x, y, z) = k$ (assuming that these extreme values exist and $\nabla g \neq \mathbf{0}$ on the surface $g(x, y, z) = k$):

(a) Find all values x , y , z , and λ such that

$$\nabla f(x, y, z) = \lambda \nabla g(x, y, z)$$

and

$$g(x, y, z) = k.$$

(b) Evaluate f at all the points (x, y, z) that result from step (a). The largest of these values is the maximum value of f ; the smallest, the minimum value of f .

Note that the above method requires you to solve a system of four equations in four unknowns.

Also note that the Method of Lagrange Multipliers also works for functions of two variables.

Example

A rectangular box without a lid is to be made from 12 m^2 of cardboard. Find the maximum volume of such a box.

Suggested Solution:

Let x , y and z denote the length, breadth and height of the box, respectively.

We need to maximize the volume

$$V(x, y, z) = xyz$$

subject to the constraint

$$g(x, y, z) = 2xz + 2yz + xy = 12.$$

By Lagrange's method we must solve:

$$yz = \lambda (2z + y) \Rightarrow xyz = \lambda (2xz + xy) \quad (1)$$

$$xz = \lambda (2z + x) \Rightarrow xyz = \lambda (2yz + xy) \quad (2)$$

$$xy = \lambda (2x + 2y) \Rightarrow xyz = \lambda (2xz + 2yz) \quad (3)$$

and

$$2xz + 2yz + xy = 12. \quad (4)$$

Now, if $\lambda = 0$, then $yz = xz = xy = 0$ which contradicts the fact that $2xz + 2yz + xy = 12$. Hence, $\lambda \neq 0$.

Moreover, we may assume that $x \neq 0$, $y \neq 0$ and $z \neq 0$; for otherwise the volume of the box is 0 which will definitely not be a maximum value.

Setting (1) and (2) equal, yields

$$\lambda(2xz + xy) = \lambda(2yz + xy) \Rightarrow xz = yz,$$

and so, $x = y$. Similarly, by setting (1) and (3) equal, we obtain

$$\lambda(2xz + xy) = \lambda(2xz + 2yz) \Rightarrow xy = 2yz,$$

and so, $z = \frac{1}{2}x$.

Substituting $x = y$ and $z = \frac{1}{2}x$ in (4) now gives

$$2x \left(\frac{1}{2}x \right) + 2(x) \left(\frac{1}{2}x \right) + x(x) = 12 \Rightarrow x^2 = 4.$$

Hence, since x cannot be negative (why?), it follows that $x = 2$.
Thus, $y = 2$ and $z = 1$.

The maximum volume is therefore given by

$$V(2, 2, 1) = 4 \text{ m}^3.$$

Example

Find the extreme values of the function $f(x, y) = x^2 + 2y^2$ on the circle $x^2 + y^2 = 1$.

Example

Find the extreme values of $f(x, y) = x^2 + 2y^2$ on the disk $x^2 + y^2 \leq 1$.

Example

Find the points on the sphere $x^2 + y^2 + z^2 = 4$ that are closest to and farthest from the point $(3, 1, -1)$.

We can extend the Method of Lagrange Multipliers to finding the maximum and minimum values of a function subject to two constraints.

The basic idea rests upon the fact that if $f(x, y, z)$ is a function subject to the constraints $g(x, y, z) = k$ and $h(x, y, z) = c$, then, assuming that the gradient vectors ∇g and ∇h are nonzero and not parallel, we may write

$$\nabla f(x, y, z) = \lambda \nabla g(x, y, z) + \mu \nabla h(x, y, z)$$

for some $\lambda, \mu \in \mathbb{R}$.

Considering the components in the previous equation we obtain the following system of simultaneous equations:

$$f_x = \lambda g_x + \mu h_x$$

$$f_y = \lambda g_y + \mu h_y$$

$$f_z = \lambda g_z + \mu h_z$$

$$g(x, y, z) = k$$

$$h(x, y, z) = c.$$

Example

- (a) Maximize $\sum_{i=1}^n x_i y_i$ subject to the constraints $\sum_{i=1}^n x_i^2 = 1$
and $\sum_{i=1}^n y_i^2 = 1$.
- (b) Set

$$x_i = \frac{a_i}{\sqrt{\sum a_j^2}} \text{ and } y_i = \frac{b_i}{\sqrt{\sum b_j^2}}$$

to show that

$$\sum a_i b_i \leq \sqrt{\sum a_j^2} \sqrt{\sum b_j^2}$$

for any numbers $a_1, \dots, a_n, b_1, \dots, b_n$.